



Autumn Examinations, 2015/2016

Exam Code(s)	3BME1, 3BMS2, 3BS9, 4MBE2, 4BMS2, 4BS2, 1HA1
Exam(s)	3rd Arts, 3rd Science, 4th Arts, 4th Science
Module Codes	MA378, MA531
Module	Numerical Analysis II
Paper No	1
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Internal Examiner(s)	Prof. Graham Ellis Dr N. Madden ★

Instructions: **Answer THREE questions**

Duration 2 Hours
Number of Pages 3 (including this page)
Discipline) Mathematics

Requirements:

Release in Exam Venue	Yes ☐	No ☒
Release to Library	Yes ☐	No ☒
Statistical/ Log Tables	Yes ☒	No ☐
Graph Paper	None	
Log Graph Paper	None	
Graphic material in colour	Yes ☐	No ☒

- Q1. (a) [11 MARKS] Let p_n be a polynomial of degree n that interpolates a function f at $n + 1$ distinct points: $\{x_0, x_1, \dots, x_n\}$ (i.e., “ p_n solves the polynomial interpolation problem”).
- Show that there is only one such p_n .
 - Give a formula for the *Lagrange polynomials* associated with the points $x_0 < x_1 < \dots < x_n$.
 - Explain carefully how these polynomials can be used to solve the polynomial interpolation problem.
- (b) [11 MARKS]
- State and prove Cauchy’s theorem for $f(x) - p_n(x)$.
 - Can this theorem be used to conclude that $p_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$? Explain your answer.
- (c) [11 MARKS] Let p_4 be the polynomial that interpolates $f(x) = \exp(-x/2)$ at the points $\{0, 1, 2, 3, 4\}$. Use Cauchy’s Theorem to give a bound for the error at $x = 3/4$. (You don’t have to give a formula for p_4).

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- Q2. (a) [11 MARKS]
- Define the *piecewise linear spline interpolant*, l , to a function f , at the interpolation points $\{x_0, x_1, \dots, x_n\}$.
 - Suppose that the interpolation points are equally spaced, and that $h = x_i - x_{i-1}$ for each $i = 1, 2, \dots, n$. Using Cauchy’s theorem, prove that
- $$\max_{x_0 \leq x \leq x_n} |f(x) - l(x)| \leq \frac{h^2}{8} \max_{x_0 \leq x \leq x_n} |f''(x)|. \quad (1)$$
- Can the bound in Equation (1) be used to conclude that $l(x) \rightarrow f(x)$ as $n \rightarrow \infty$? Explain your answer.
- (b) [11 MARKS] Take $f = 2x \exp(x - 1)$, $x_0 = -1$ and $x_n = 1$.
- Write down a formula for the linear spline interpolant to f when $n = 2$.
 - Give an upper bound for the maximum error on $[-1, 1]$ for the linear spline interpolant to f with $n = 2$.
 - What is the smallest n_* one can choose so that $\max_{-1 \leq x \leq 1} |f(x) - l(x)| \leq 10^{-6}$ for all $n \geq n_*$?
- (c) [11 MARKS] Let $\hat{l}$ be *any* piecewise linear approximation to a given function f on a fixed set of interpolation points, and l be the piecewise linear interpolant to f . Show that

$$\max_{x_0 \leq x \leq x_n} |f(x) - l(x)| \leq 2 \max_{x_0 \leq x \leq x_n} |f(x) - \hat{l}(x)|.$$

What is the significance of this result?

Q3. (a) [11 MARKS] What is meant by the *precision* of a quadrature method?
Determine the precision of the following schemes for estimating $\int_{-1}^1 f(x)dx$.

- (i) $Q(f) = 2f(0)$.
- (ii) $Q(f) = (f(-1) + 4f(0) + f(1))/3$.

(b) [11 MARKS]

- (i) Explain *Gaussian* Quadrature.
- (ii) Derive the two-point Gaussian Quadrature rule of approximating $\int_{-1}^1 f(x)dx$.

(c) [11 MARKS] Let $\{\hat{p}_n(x)\}_{n=0}^\infty$ be the sequence of orthogonal monic polynomials with respect to the inner product

$$(f, g) := \int_a^b f(x)g(x)dx.$$

- (i) Prove that $\hat{p}_n(x)$ is orthogonal to *all* polynomials of degree less than n .
- (ii) Prove that the zeros of the $\hat{p}_n(x)$ are simple (i.e., distinct).
- (iii) Prove that the zeros of the $\hat{p}_n(x)$ are all in the interval $[a, b]$.

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Q4. (a) [12 MARKS] Let u be the solution to the differential equation

$$-u''(x) + r(x)u(x) = e^{-x} \quad \text{on} \quad (0, 1) \quad \text{and} \quad u(0) = u(1) = 0. \quad (2)$$

Let ϱ be such $r(x) \geq \varrho > 0$. Prove that, for any $x \in [0, 1]$,

$$0 \leq u(x) \leq 1/\varrho.$$

(b) [6 MARKS] Write down the variational (i.e., weak) form of the differential equation (2). Show that the solution to this variational problem is unique.

(c) [10 MARKS] Use the finite element method with the usual piecewise linear Galerkin basis functions (i.e., “hat” functions) on the mesh $\{0, 1/3, 2/3, 1\}$ to find an approximate solution to

$$-u''(x) = 3 \quad \text{on} \quad (0, 1) \quad \text{and} \quad u(0) = u(1) = 0.$$

(d) [5 MARKS] State Céa’s lemma, and explain its relevance to error analysis of the finite element method.